JALRAKSHAK AI

Flood & Drought Early Warning System

Research-enhanced proposal for Neural Storm 2026

Prepared by Ashish

Project concept: A geospatial decision-support prototype combining environmental observations and baseline machine-learning models to estimate localized flood and drought risk in selected Uttar Pradesh pilot areas.

Recommended pilot design

PILOT	WHY IT FITS	WHAT WE WILL TEST
Gorakhpur Flood-risk pilot	UPSDMA's 2024 flood mock-exercise report lists Gorakhpur among hyper-sensitive flood districts and within the Rapti basin's affected districts.	Rainfall features plus credible historical flood / inundation labels, if obtainable.
Mahoba Drought-risk pilot	Mahoba is in Bundelkhand, a region relevant to drought risk. Confirm the exact drought classification and labels from official sources before claiming model training readiness.	Seasonal rainfall deficits and dry spells; add vegetation / soil-moisture variables only if accessible and quality-checked.

What makes this proposal stronger

- Clear distinction between an environmental risk estimate and an official early warning.
- Specific, documented data sources and honest spatial-resolution limits.
- Two separate hazard models with explicit target labels and temporal evaluation.
- A compact six-slide narrative designed for a fast selection-round review.

Core credibility rule: Do not claim model accuracy, forecast lead time, or live alerts until those functions have been implemented and tested.

SLIDE 1 — TITLE & VISION

JalRakshak AI: Environmental Data into Risk Intelligence

Slide copy: AI-powered geospatial risk assessment for a more disaster-resilient Uttar Pradesh. Neural Storm 2026 — Python AI/ML Hackathon.

Visual direction

- Use one strong split visual: floodwater / river basin on one side, drought-stressed farmland on the other.
- Add a simple map pin or satellite-grid motif; avoid a busy collage.

Speaker note

Introduce this as a software prototype that turns environmental observations into location-specific risk estimates and preparedness information—not a system that guarantees disaster prediction.

SLIDE 2 — PROBLEM

Floods and Droughts Need Different Signals

Flood challenge: Heavy rainfall, river conditions, terrain and embankment issues can contribute to inundation. Uttar Pradesh's official flood action plan describes the state as flood-prone and notes the importance of preparedness and district-level coordination. [1][2]

Drought challenge: Rainfall deficits can accumulate over weeks or months and affect soil moisture, vegetation, crops and water availability. Drought assessment requires seasonal context rather than a single day's weather.

The gap: People need environmental data converted into an understandable local risk picture—with the data date, reasons for the estimate, and links to official warnings clearly shown.

Avoid

- Do not use unsupported loss totals or current disaster statistics without checking their date and source.
- Do not imply a generic chatbot or a weather dashboard alone is a predictive ML system.

SLIDE 3 — SOLUTION & PILOT AREA

One Platform, Two Hazard-Specific Modules

Slide copy

- Flood-risk assessment pilot: Gorakhpur, with Rapti basin context.
- Drought-risk assessment pilot: Mahoba, within the Bundelkhand context; validate the exact drought labels before model training.
- Interactive map, environmental trend charts, risk category, explanation of contributing factors, and official alert links.

Why Gorakhpur?

UPSDMA's 2024 flood mock-exercise report lists Gorakhpur among the state's hyper-sensitive flood districts and identifies it among affected districts in the Rapti River basin. [1]

Why Mahoba?

Mahoba is a plausible drought-focused pilot in Bundelkhand, but verify the exact district-level drought designation and labels in official planning documents before asserting that it is the most drought-vulnerable district. [2][3]

Selection-round message: Start with two small, hazard-specific pilots rather than claiming state-wide coverage from the first version.

What users see

- Selected district and data timestamp.
- Flood / drought module selected.
- Risk category plus the indicators used by the model.
- Cautious preparedness guidance and links to official sources.

SLIDE 4 — DATA & AI/ML

Real Data + Two Baseline Models

Data sources (verify access path and version in implementation)

SOURCE	ROLE	ACCURACY / LIMIT NOTE
NASA GPM IMERG [4][5]	Satellite precipitation estimates; useful for rainfall accumulations and event context.	0.1° × 0.1° grid (~10 km scale); products can be 30-minute or daily. It is not street-level rainfall measurement.
CHIRPS daily rainfall [6]	Long historical rainfall series for seasonal comparisons, rainfall anomalies and dry-spell features.	Daily gridded rainfall; document product version, units, period and geographic aggregation.
Official / satellite event labels	Flood-event / inundation labels and a defined drought index or documented drought outcomes.	Rainfall alone is not a flood label. Choose labels that are aligned by date and area.
Terrain / elevation	Geographic context for flood susceptibility.	Terrain alone cannot establish flood occurrence or depth.

Flood baseline

- Features: multi-day rainfall accumulation / intensity; terrain; available river context; historical flood or inundation labels.
- Algorithm: Random Forest classifier if reliable labels are obtained. If labels are weak, call it a preliminary susceptibility / heavy-rainfall risk proxy.

Drought baseline

- Features: rainfall deficit against historical seasonal conditions, dry-spell duration, and optional vegetation / soil-moisture signals.
- Algorithm: Random Forest trained against a clearly defined drought index / event label. State whether the target is meteorological or agricultural drought.

Evaluation

- Use a chronological train/test split; avoid random mixing that leaks information from the future.
- Report precision, recall, false alarms, missed events, and confusion matrix. Put actual numbers in the PPT only after running evaluation.

Do not call an arbitrary 0–100 score a probability. A probability claim requires calibration and validation.

SLIDE 5 — ARCHITECTURE & MVP

Software-Only Architecture

LAYER	TOOLS	JOB
Data pipeline	Python, pandas; geospatial libraries if needed	Download, clean, align dates / regions, build features and save data provenance.
ML models	scikit-learn Random Forest baselines	Separate flood and drought model training, inference and evaluation.
API	FastAPI	Serve district-level indicators, model output and timestamp.
Frontend	React / Next.js, Leaflet, charts	Map, risk cards, trends, factors and source links.
Optional LLM	Gemma / other language model	Explain structured model output; never invent predictions or issue an unverified warning.

Demo flow

- 1. Select Gorakhpur or Mahoba.
- 2. Show available environmental observations and their timestamp.
- 3. Run the corresponding baseline model on the correct features.
- 4. Display risk category, contributing factors and limitations.
- 5. Compare a historical case with documented outcomes, where labels are available.

Build order

- First: obtain data and labels. Second: train/evaluate model. Third: connect API. Fourth: build dashboard. Fifth: test and rehearse.

No physical sensors are required for the MVP. Optional future work could integrate river gauges, rain gauges, or soil-moisture sensors.

SLIDE 6 — IMPACT & RESPONSIBLE USE

A Practical Pilot with a Path to Scale

Expected value

- Make local environmental indicators easier to inspect in one map-based interface.
- Support preparedness discussions for farmers, residents and disaster-management teams.
- Provide a reusable baseline that can expand to more districts only after validation.

MVP milestones

- Prepare the two pilot datasets and define labels.
- Train and evaluate both baseline models.
- Integrate results into a map dashboard with source timestamps.
- Demonstrate at least one historical test case and disclose metrics / limitations.

Safety and limitations

- This is a research prototype, not an official warning service.
- Forecast lead time depends on forecast inputs and validation; rainfall observations alone do not create advance prediction.
- A district-wide average may hide local conditions; clearly show geographic resolution.
- Use official warnings as authoritative sources for emergency action.

Final pitch: JalRakshak AI combines real environmental data, hazard-specific baseline ML, and geospatial visualization to make local flood and drought risk easier to understand—without overstating forecast certainty.

Sources for slide footnotes / verification

[1] Uttar Pradesh State Flood Mock Exercise Report 2024 — Rapti River Basin and flood-sensitive districts: https://sdma.up.gov.in/assets/uploads/hydrological/1747389914_flood_me_report_2024%20%281%29.pdf

[2] UPSDMA Hazard Profile of Uttar Pradesh: https://upsdma.up.nic.in/stateprofile.htm

[3] UPSDMA Flood Action Plan 2023 and official hazard resources: https://upsdma.up.nic.in/2023/Flood-Action-Plan-2023.pdf ; https://upsdma.up.nic.in/

[4] NASA GPM IMERG data page: https://gpm.nasa.gov/data/imerg

[5] NASA GPM data directory: https://gpm.nasa.gov/data/directory

[6] CHIRPS daily dataset catalog: https://developers.google.com/earth-engine/datasets/catalog/UCSB-CHG_CHIRPS_DAILY

Canva guidance: One key message per slide; use 3–5 short bullets, one diagram or map, and source footnotes in small type. Put detail in speaker notes. Replace any proposed capability with a truthful statement of what has actually been implemented.